

*Achievements of School Educational Data  
Analysis With SQL & Power BI*

**By Hassan Nezamoleslam**

**In this study, multiple visualisations were developed to support data-driven analysis of student performance.**

**These outputs can be used in meetings with parents, teachers, and academic advisors to evaluate students quantitatively, rather than relying solely on qualitative assessments.**

**1. Analysis of average grades across all students for each subject in both mock exams and final exams. This enables tracking of performance trends (improvement or decline) and evaluation of the reliability of mock exams.**

**2. A more detailed analysis can be conducted at the individual student level, allowing performance tracking for each student separately.**

**3. A comprehensive overview of each student's performance across all subjects can be monitored, providing a deeper level of insight compared to the previous stage.**

**4. Identification of top-performing and underperforming students in each subject, enabling tailored academic strategies and learning pathways.**

**5. Simultaneous monitoring of an individual student's performance using multiple visualisations, along with comparison against peers.**

**6. Evaluation of each student's performance relative to the class average, highlighting whether they are above or below the overall level.**

**7. Ranking of average grades across subjects and student groups (e.g., by class or gender), enabling better understanding of performance patterns and fostering healthy academic competition.**

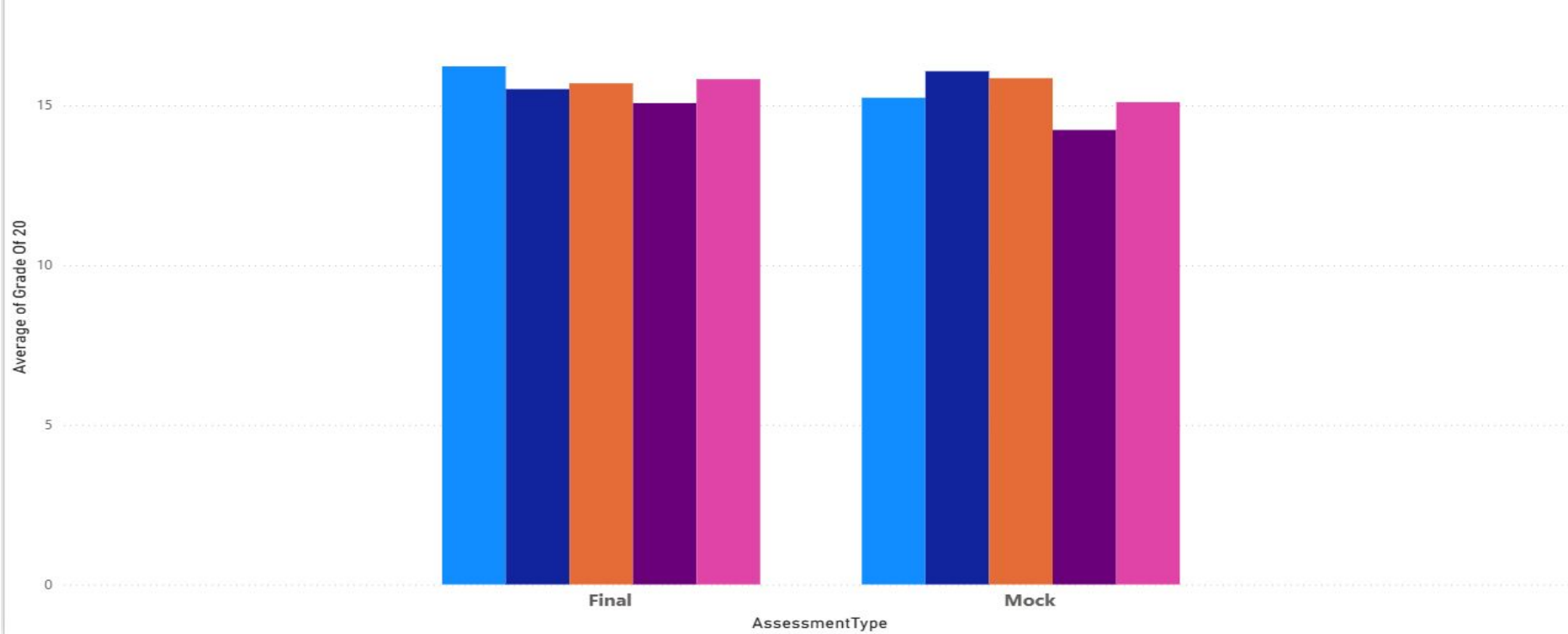
**These visualisations and findings represent a simplified sample. More advanced scenarios and analyses can be developed to extend the scope and depth of insights.**

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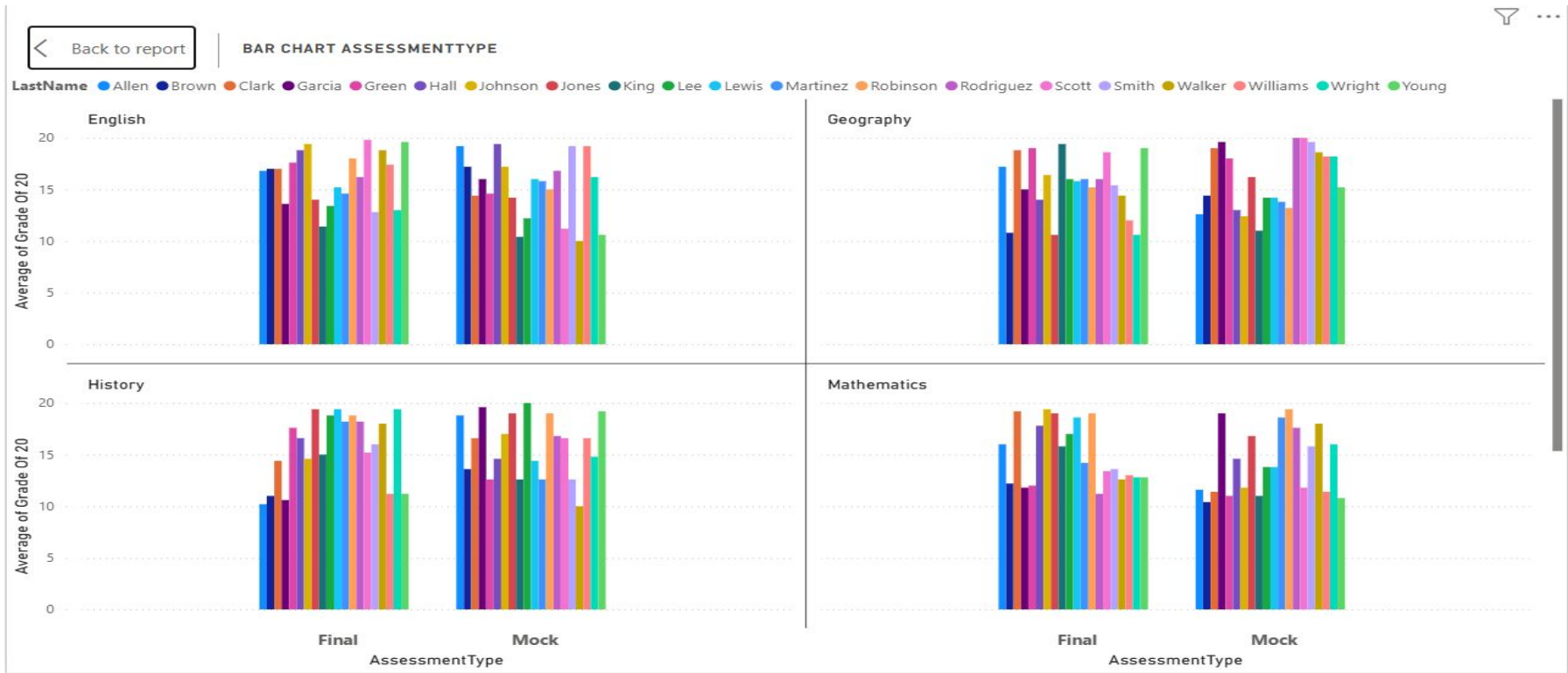
GARDES (MOCK +FINAL) BY SUBJECTS



SubjectName ● English ● Geography ● History ● Mathematics ● Science



*Average Grades & AssessmentType(Mock,Final), Subject Legend*



*Grades & SubjectName For Every Student*

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## MATRIX OF EXAM

| AssessmentType | Allen        | Brown        | Clark        | Garcia       | Green        | Hall         | Johnson      | Jones        | King         | Lee          | Lewis        | Martinez     | Robinson     | Rodriguez    | Scott        | Smith        | Walker       | Williams     | Wright       | Young        | Total        |
|----------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| <b>Final</b>   | <b>14.48</b> | <b>13.00</b> | <b>17.60</b> | <b>13.08</b> | <b>15.96</b> | <b>16.72</b> | <b>17.36</b> | <b>15.96</b> | <b>14.32</b> | <b>15.32</b> | <b>17.72</b> | <b>15.88</b> | <b>18.00</b> | <b>15.80</b> | <b>16.00</b> | <b>15.08</b> | <b>16.36</b> | <b>13.88</b> | <b>14.84</b> | <b>15.88</b> | <b>15.66</b> |
| English        | 16.80        | 17.00        | 17.00        | 13.60        | 17.60        | 18.80        | 19.40        | 14.00        | 11.40        | 13.40        | 15.20        | 14.60        | 18.00        | 16.20        | 19.80        | 12.80        | 18.80        | 17.40        | 13.00        | 19.60        | <b>16.22</b> |
| Geography      | 17.20        | 10.80        | 18.80        | 15.00        | 19.00        | 14.00        | 16.40        | 10.60        | 19.40        | 16.00        | 15.80        | 16.00        | 15.20        | 16.00        | 18.60        | 15.40        | 14.40        | 12.00        | 10.60        | 19.00        | <b>15.51</b> |
| History        | 10.20        | 11.00        | 14.40        | 10.60        | 17.60        | 16.60        | 14.60        | 19.40        | 15.00        | 18.80        | 19.40        | 18.20        | 18.80        | 18.20        | 15.20        | 16.00        | 18.00        | 11.20        | 19.40        | 11.20        | <b>15.69</b> |
| Mathematics    | 16.00        | 12.20        | 19.20        | 11.80        | 12.00        | 17.80        | 19.40        | 19.00        | 15.80        | 17.00        | 18.60        | 14.20        | 19.00        | 11.20        | 13.40        | 13.60        | 12.60        | 13.00        | 12.80        | 12.80        | <b>15.07</b> |
| Science        | 12.20        | 14.00        | 18.60        | 14.40        | 13.60        | 16.40        | 17.00        | 16.80        | 10.00        | 11.40        | 19.60        | 16.40        | 19.00        | 17.40        | 13.00        | 17.60        | 18.00        | 15.80        | 18.40        | 16.80        | <b>15.82</b> |
| <b>Mock</b>    | <b>14.60</b> | <b>14.20</b> | <b>15.08</b> | <b>18.68</b> | <b>13.48</b> | <b>15.04</b> | <b>14.20</b> | <b>16.48</b> | <b>12.32</b> | <b>15.28</b> | <b>15.44</b> | <b>15.52</b> | <b>15.80</b> | <b>18.20</b> | <b>15.48</b> | <b>16.12</b> | <b>15.08</b> | <b>16.04</b> | <b>15.60</b> | <b>13.32</b> | <b>15.30</b> |
| English        | 19.20        | 17.20        | 14.40        | 16.00        | 14.60        | 19.40        | 17.20        | 14.20        | 10.40        | 12.20        | 16.00        | 15.80        | 15.00        | 16.80        | 11.20        | 19.20        | 10.00        | 19.20        | 16.20        | 10.60        | <b>15.24</b> |
| Geography      | 12.60        | 14.40        | 19.00        | 19.60        | 18.00        | 13.00        | 12.40        | 16.20        | 11.00        | 14.20        | 14.20        | 13.80        | 13.20        | 20.00        | 20.00        | 19.60        | 18.60        | 18.20        | 18.20        | 15.20        | <b>16.07</b> |
| History        | 18.80        | 13.60        | 16.60        | 19.60        | 12.60        | 14.60        | 17.00        | 19.00        | 12.60        | 20.00        | 14.40        | 12.60        | 19.00        | 16.80        | 16.60        | 12.60        | 10.00        | 16.60        | 14.80        | 19.20        | <b>15.85</b> |
| Mathematics    | 11.60        | 10.40        | 11.40        | 19.00        | 11.00        | 14.60        | 11.80        | 16.80        | 11.00        | 13.80        | 13.80        | 18.60        | 19.40        | 17.60        | 11.80        | 15.80        | 18.00        | 11.40        | 16.00        | 10.80        | <b>14.23</b> |
| Science        | 10.80        | 15.40        | 14.00        | 19.20        | 11.20        | 13.60        | 12.60        | 16.20        | 16.60        | 16.20        | 18.80        | 16.80        | 12.40        | 19.80        | 17.80        | 13.40        | 18.80        | 14.80        | 12.80        | 10.80        | <b>15.10</b> |
| <b>Total</b>   | <b>14.54</b> | <b>13.60</b> | <b>16.34</b> | <b>15.88</b> | <b>14.72</b> | <b>15.88</b> | <b>15.78</b> | <b>16.22</b> | <b>13.32</b> | <b>15.30</b> | <b>16.58</b> | <b>15.70</b> | <b>16.90</b> | <b>17.00</b> | <b>15.74</b> | <b>15.60</b> | <b>15.72</b> | <b>14.96</b> | <b>15.22</b> | <b>14.60</b> | <b>15.48</b> |

***Fast Look Of Student's Situation***

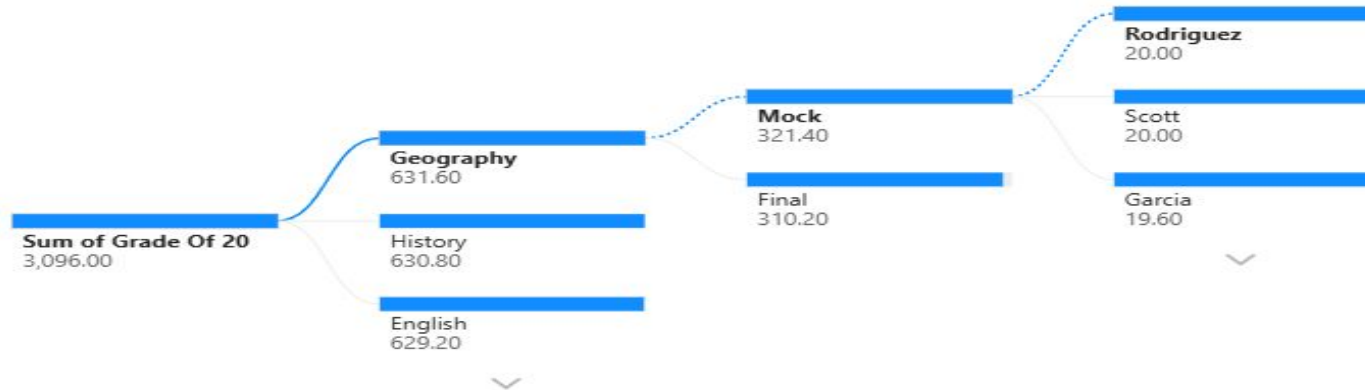
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## THE FIVE TOP STUDENTS

**SubjectName** ×  
Geography

**AssessmentType** ×  
Mock

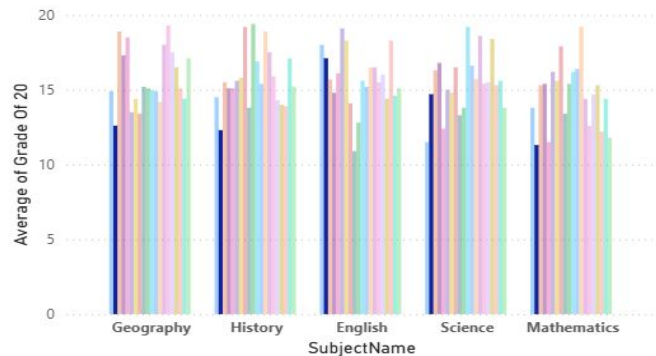
**LastName** ×



***Rank Of Top Students Base on Subject And AssessmentType***

## Student's Grades

LastName ● Allen ● Brown ● Clark ● Garcia ● Green ● Hall ● Johnson



## Ask Me

⚠ Q&A will be retiring in December 2026.

Remove Q&A

Grace (JoinAcademicGradesSubjectStudent (2) first name)

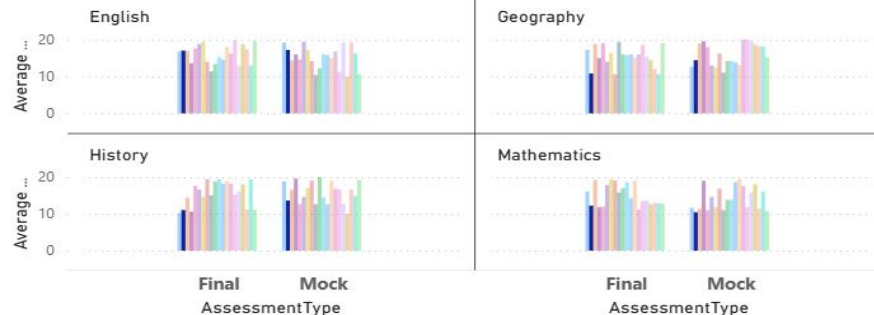
StudentID AssessmentDate AssessmentType FirstName LastName Score

## Matrix Of Exam

| AssessmentType | Allen        | Brown        | Clark        | Garcia       | Green        | Hall         | Johnson      | Jones        | King         | Lee          | Lewis        |
|----------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| <b>Final</b>   | <b>14.48</b> | <b>13.00</b> | <b>17.60</b> | <b>13.08</b> | <b>15.96</b> | <b>16.72</b> | <b>17.36</b> | <b>15.96</b> | <b>14.32</b> | <b>15.32</b> | <b>17.72</b> |
| English        | 16.80        | 17.00        | 17.00        | 13.60        | 17.60        | 18.80        | 19.40        | 14.00        | 11.40        | 13.40        | 15.20        |
| Geography      | 17.20        | 10.80        | 18.80        | 15.00        | 19.00        | 14.00        | 16.40        | 10.60        | 19.40        | 16.00        | 15.80        |
| History        | 10.20        | 11.00        | 14.40        | 10.60        | 17.60        | 16.60        | 14.60        | 19.40        | 15.00        | 18.80        | 19.40        |
| Mathematics    | 16.00        | 12.20        | 19.20        | 11.80        | 12.00        | 17.80        | 19.40        | 19.00        | 15.80        | 17.00        | 18.60        |
| Science        | 12.20        | 14.00        | 18.60        | 14.40        | 13.60        | 16.40        | 17.00        | 16.80        | 10.00        | 11.40        | 19.60        |
| <b>Mock</b>    | <b>14.60</b> | <b>14.20</b> | <b>15.08</b> | <b>18.68</b> | <b>13.48</b> | <b>15.04</b> | <b>14.20</b> | <b>16.48</b> | <b>12.32</b> | <b>15.28</b> | <b>15.44</b> |
| English        | 19.20        | 17.20        | 14.40        | 16.00        | 14.60        | 19.40        | 17.20        | 14.20        | 10.40        | 12.20        | 16.00        |
| Geography      | 12.60        | 14.40        | 19.00        | 19.60        | 18.00        | 13.00        | 12.40        | 16.20        | 11.00        | 14.20        | 14.20        |
| History        | 18.80        | 13.60        | 16.60        | 19.60        | 12.60        | 14.60        | 17.00        | 19.00        | 12.60        | 20.00        | 14.40        |
| Mathematics    | 11.60        | 10.40        | 11.40        | 19.00        | 11.00        | 14.60        | 11.80        | 16.80        | 11.00        | 13.80        | 13.80        |
| Science        | 10.80        | 15.40        | 14.00        | 19.20        | 11.20        | 13.60        | 12.60        | 16.20        | 16.60        | 16.20        | 18.80        |
| <b>Total</b>   | <b>14.54</b> | <b>13.60</b> | <b>16.34</b> | <b>15.88</b> | <b>14.72</b> | <b>15.88</b> | <b>15.78</b> | <b>16.22</b> | <b>13.32</b> | <b>15.30</b> | <b>16.58</b> |

## Bar Chart AssessmentType

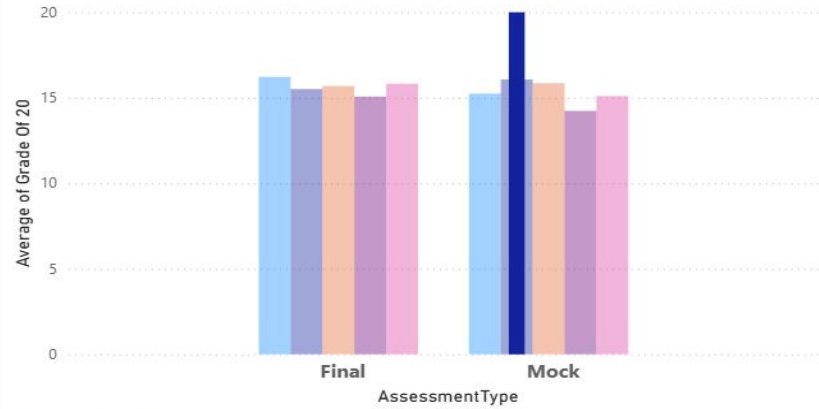
LastName ● Allen ● Brown ● Clark ● Garcia ● Green ● Hall ● Johnson ● Jones ● King ● Lee ● Lewis



***Trace Student's Situation In the different charts Simultaneously***

## Gardes (Mock +Final) By Subjects

SubjectName ● English ● Geography ● History ● Mathematics ● Science



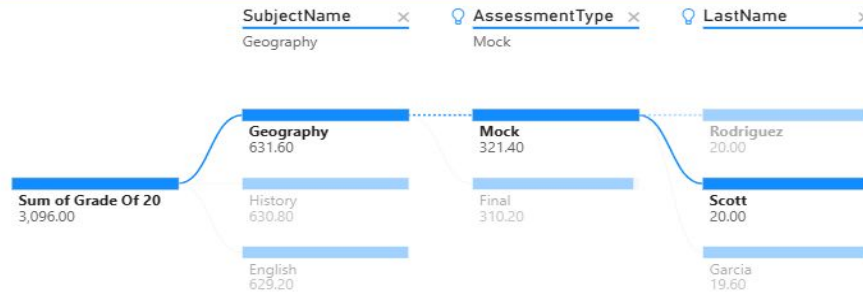
## Quick Look Of The Exams

| AssessmentType | Geography | Total |
|----------------|-----------|-------|
| Mock           | 20.00     | 20.00 |
| Total          | 20.00     | 20.00 |

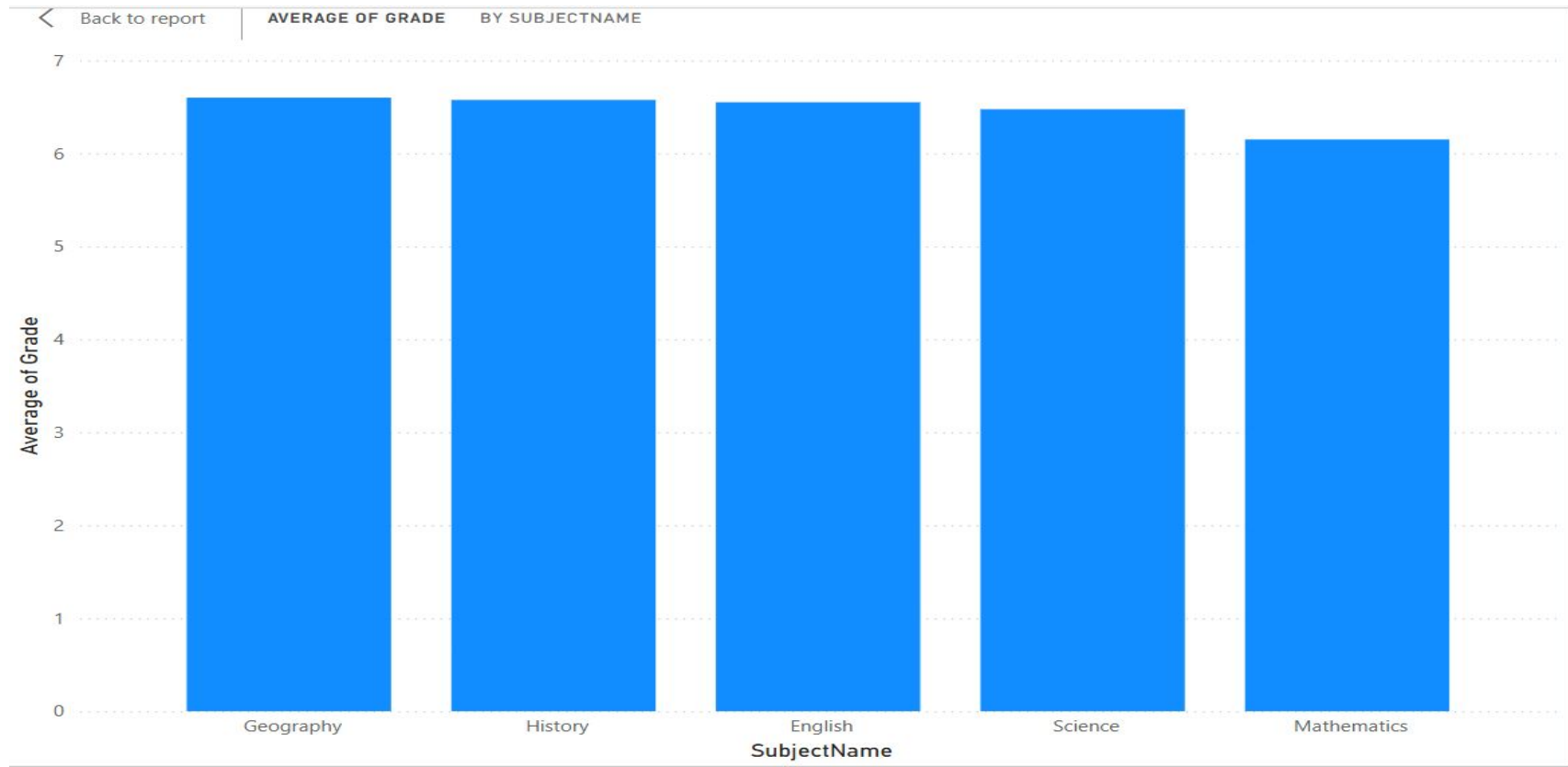
## Analysis Of AssessmentType



## The Five Top Students



***Compare A Student's Grades with The Average***



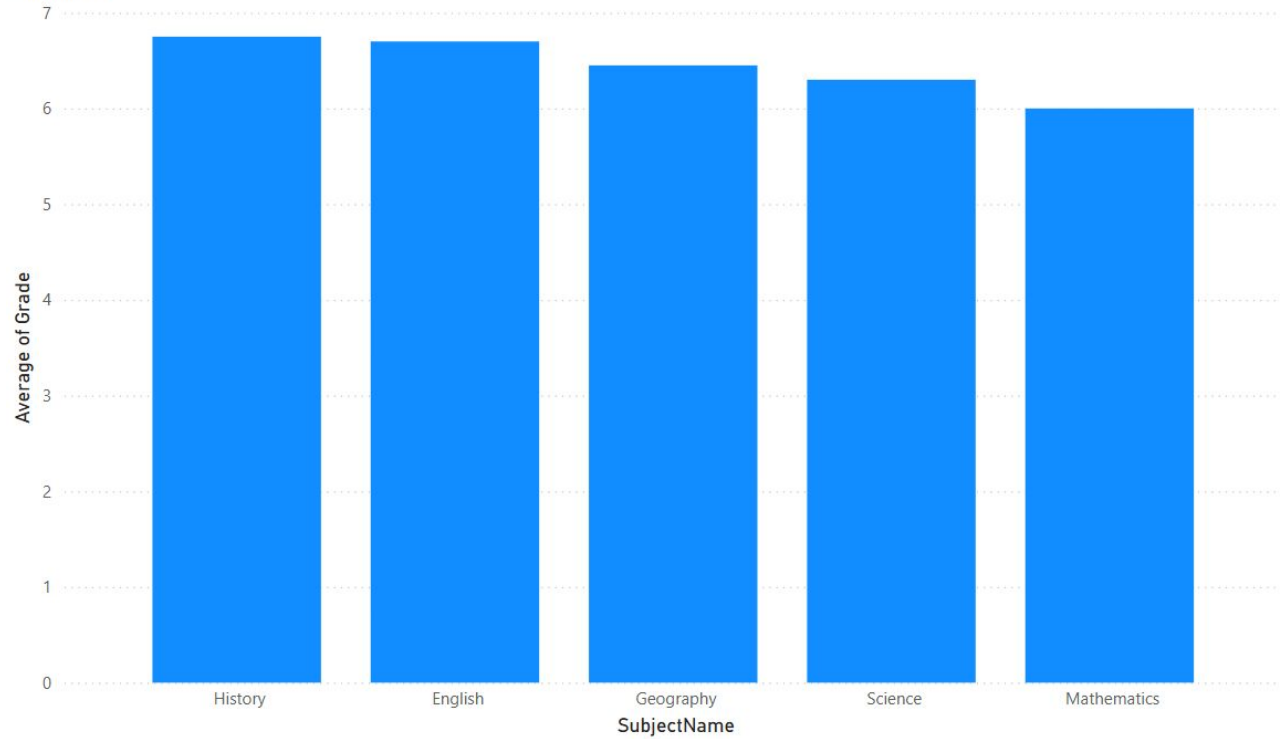
***Sort The Average Grades For Different SubjectName***

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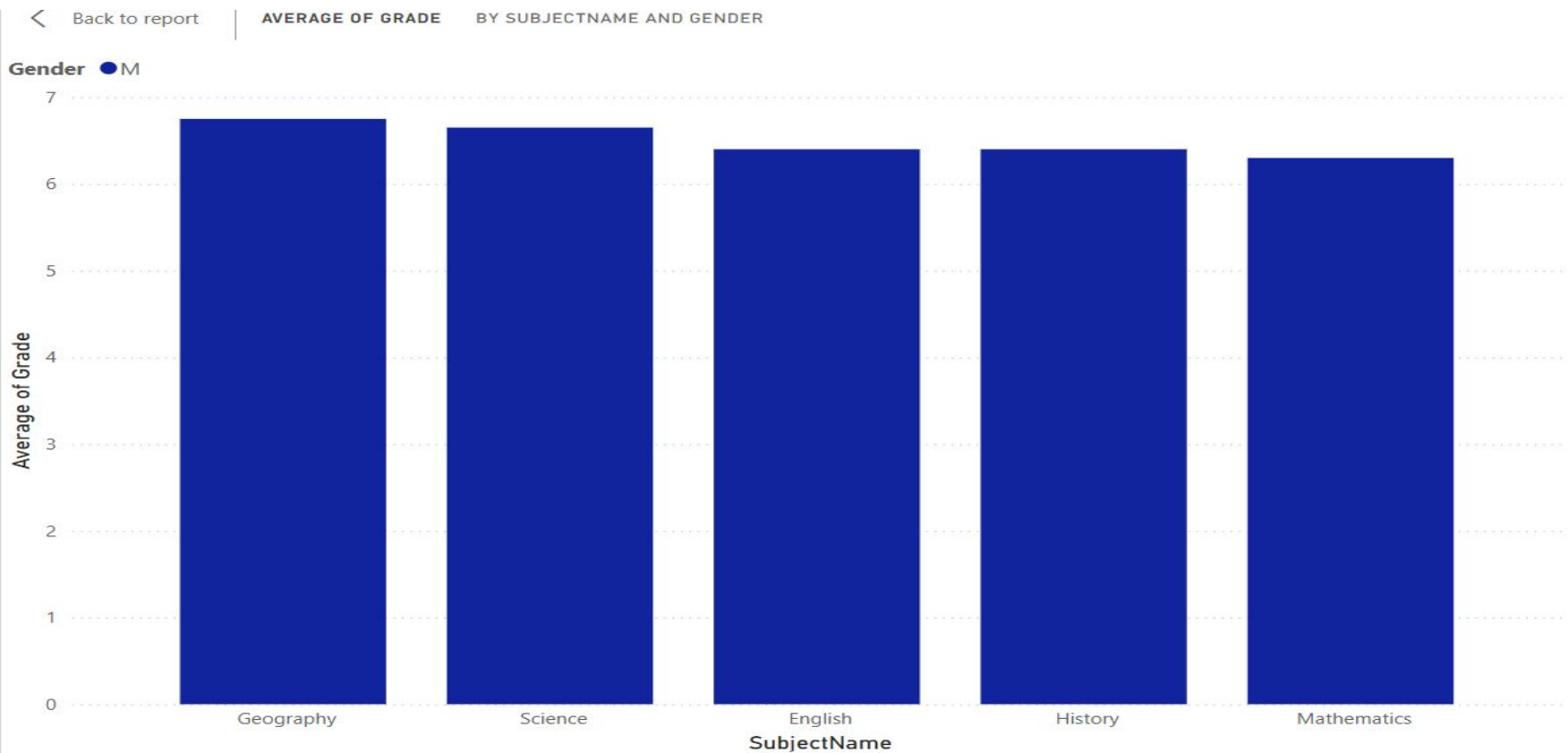
AVERAGE OF GRADE

BY SUBJECTNAME AND GENDER

Gender ● F



***Sort The Average Grades Base On SubjectName&Gender 'Girls'***



***Sort The Average Grades Base On SubjectName&Gender 'Boys'***